USN | | | | | | | | | | [20EEOE02]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2023

OPERATIONS RESEARCH

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.
- **iii.** Normal distribution table will be provided

1. a. List out various phases of Operation research and explain in brief all of them. **07**

b. Solve the following linear programming problem graphically: **07**

Maximise $Z = 4x + y$ ... (1)

subject to the constraints:

$x + y \leq 50$ ... (2)

$3x + y \leq 90$ ... (3)

$x \geq 0, y \geq 0$ ... (4)

2. a. A dairy firm has three plants located in a state. The daily milk production at each plant is as follows: **08**

Plant 1 : 6 million litres, Plant 2 : 1 million litres, and Plant 3 : 10 million litres

Each day, the firm must fulfil the needs of its four distribution centres.

The minimum requirement of each centre is as follows:

Distribution centre 1 : 7 million litres, Distribution centre 2 : 5 million litres, Distribution centre 3 : 3 million litres, and Distribution centre 4 : 2 million litres

Cost (in hundreds of rupees) of shipping one million litre from each plant to each distribution centre is given in the following table:

		Distribution Centre			
		D_1	D_2	D_3	D_4
Plant	P_1	2	3	11	7
	P_2	1	0	6	1
	P_3	5	8	15	9

Find the initial basic feasible solution for given problem by using following methods:

- **i.** North-west corner rule
- **ii.** Least cost method
- **iii.** Vogel's approximation method

b. In a computer centre after studying carefully the three expert programmes, the head of computer centre, estimates the computer time in minutes required by the experts for the application programmes as follows: **06**

		Programmes		
		A	B	C
Programmes	1.	120	100	80
	2.	80	90	110
	3.	110	140	120

Assign the programmers to the programmes in such a way that the total computer time is minimum.

3. **a.** Define Saddle point, Zero Sum game, Game Value. **06**

b. Find Solution of game theory problem using dominance and graphical method **08**

Player A\Player B	B1	B2	B3	B4
A1	2	2	3	-2
A2	4	3	2	6

OR

4. **a.** Explain maximin – minimax principle. **04**

b. Find Solution of game theory problem using saddle point **05**

Player A\Player B	B1	B2	B3	B4
A1	20	15	12	35
A2	25	14	8	10
A3	40	2	10	5
A4	-5	4	11	0

c. In a certain game, player A has three possible choices L, M and N while player B has two possible choices P and Q. payments are to be made according to the choices made **05**

Choices	Payment
L, P	A pays B Rs 3
L, Q	B pays A Rs 3
M, P	A pays B Rs 2
M, Q	B pays A Rs 4
N, P	B pays A Rs 2
N, Q	B pays A Rs 3

What are the best strategies for players A and B in this game? What is the value of the game for A and B?

5. **a.** A Project is composed of seven activities whose time estimates are listed in the following table. Activities are simplified by this beginning (1) ones ending (j) Node member. Calculate expected project length. **09**

Activity		Estimated duration in weeks		
i	*j*	*Optimistic*	*Most likely*	*Pessimestic*
1	2	1	1	7
1	3	1	4	7
1	4	2	2	8
2	5	1	1	1
3	5	2	5	14
4	6	2	5	8
5	6	3	6	15

b. Define a project. Explain the basic steps in project management. **05**

OR

6. a. Explain the steps involved in Fulkerson's rule (i - j) rule. **04**

b. A project schedule has the following characteristics: **10**

Activity	Time
1 – 2	4
1 – 3	1
2 – 4	1
3 – 4	1
3 – 5	6
4 – 9	5
5 – 6	4
5 – 7	8
6 – 8	1
7 – 8	2
8 – 10	5
9 – 10	7

i. Construct network diagram

ii. Compute earliest and latest times of each event

iii. Calculate the floating times

iv. Find the critical path and project duration

7. a. What are the types of demand pattern? Explain them with suitable sketches **05**

b. A company adopts method of least squares to develop a linear trend equation for the data as shown in the table below. **09**

YEAR(X)	1	2	3	4	5	6	7	8	9	10	11
Shipment in tones (y)	2	3	6	10	8	7	12	14	14	18	19

Calculate the trend forecast for the year 12 and 20.

OR

8. a. Define forecasting. Classify and explain the different types of forecasting **08**

b. Discuss the steps of Delphi method **06**